

AI Voice-Based User Interface (VUI) for Hands-Free Interaction

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Overview and Motivation

This technique is called a Voice-Based User Interface (VUI). It allows users to interact with a system using their voice instead of typing or clicking. The main goal is to make interaction faster and more natural. With recent AI improvements, voice systems are now more accurate and actually usable in real situations. This matters especially for people who cannot use their hands easily or are in situations where typing is not practical.

Target Users and Use Cases

The main users include people with physical disabilities, workers who need to multitask, and everyday users who want faster interaction. This technique is used in things like virtual assistants, workplace tools, and accessibility systems. It is especially useful in environments where hands-free interaction saves time or effort.

How the Technique Works

The system uses AI to convert speech into text in real time. Then it processes the meaning of the sentence and turns it into an action. For example, if a user says "open a document," the system recognizes the speech and executes the command. The system improves performance by processing audio in smaller chunks and optimizing how data is handled.

Design Features and Key Insight

Important features include real-time processing, high accuracy, and the ability to understand natural language. The key idea is that speaking is more natural than typing, so using voice reduces effort and can make systems easier to use.

Evaluation and Findings

The system was evaluated using accuracy-based metrics like word error rate and similarity measures. Results showed high accuracy (around 88%) and improved efficiency when compared to manual input methods. It also showed that smaller audio processing chunks improved performance.

Assessment

This technique is strong because it allows hands-free interaction and improves accessibility. It is also faster in many cases. However, it can struggle in noisy environments and may not always understand accents or unclear speech. Overall, it is a useful and growing AI-based interaction technique.

Figure 1. Example of a voice-based interaction system where user speech is converted into commands.

Reference

Harihar, D. et al. (2025). Voice-Based User Interface for Hands-Free Data Entry and Automation at Workplaces. ScienceDirect.